

A QUEUEING-THEORETIC FRAMEWORK FOR CONTROLLING CUSTOMER FLOW IN LOYALTY PROGRAMS UNDER BUDGET CONSTRAINTS

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In this work, we contribute to queueing theory by applying queueing-theoretic methods as an economic management tool aimed at controlling customer behavior within retail loyalty programs considering system economic state.

Our framework models each loyalty level as a finite-capacity service station within a queueing network. Here, each level represents a section of the customer–product matrix that reflects the ability of customers who have achieved certain shopping milestones to purchase certain products at a discount. Each level has capacity constraints reflecting discount budgets or eligibility limits. Customers, modeled as jobs arriving stochastically (e.g., via a Poisson process with rate λ), are processed sequentially through these levels. The progression from level A_i to level A_{i+1} is governed by the state-update equation:

$$x_{i+1}(t+1) = x_{i+1}(t) + u_i(t) \cdot \mu_i x_i(t) - D_{i+1}(t), \quad (1)$$

where $x_i(t)$ denotes the number of customers at level A_i ; $u_i(t) \in \{0, 1\}$ is a control variable indicating admission to the next level; μ_i is the progression rate (the scaling factor determining the size of the delta from $x_i(t)$); $D_{i+1}(t)$ captures departures from level A_{i+1} . The model enforces capacity constraints:

$$x_i(t) \leq C_i, \quad \forall i, t, \quad (2)$$

which effectively block further transitions when a level is full. This blocking mechanism accounts for additional economic information about the system state and serves as the control lever to regulate customer flow and prevent over-discounting.

By extending the state-dependent F-policy approach, our framework provides a systematic method for dynamically controlling customer progression. Its goal is to optimize key financial metrics by aligning discount allocation with customer behavior, ensuring that discount strategies remain both attractive and financially sustainable.